One time per channel: how lateral charge spreading biases track angles in time-projection readouts, and what to do about it

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Abstract

A micro time-projection chamber (μTPC) reconstructs a short track segment from the positions of its readout channels and one drift time per channel. Since the first Micromegas and GEM implementations the reconstructed angle has been reported too large, most strongly for tracks near normal incidence; the explanation in print is capacitive coupling between neighbouring strips, and the remedies are edge-strip filters, a subtraction of a tuned fraction of the neighbour pulse, or an angle calibration. We review this literature across detector families, from strip μTPCs to large pad TPCs, liquid-argon TPCs and directional recoil detectors, and show that one mechanism covers all of it: a single time per channel is a functional of the charge from every drift depth that reaches the channel, and any lateral spread of that charge (drift diffusion, the induction footprint, capacitive coupling, resistive spreading) feeds the strips at the two ends of the column with charge of the wrong depth. For a straight fit through N channels the slope error is a strip-end term, $\tan\theta_{\mathrm{rec}}/\tan\theta - 1 \simeq C/N^2$, plus a small interior term from the growth of the spread with depth; C is set by the lateral spread in units of the pitch and by the time estimator. We test this on a 28 mm resistive-strip Micromegas with a 0.78 mm pitch, 60 ns sampling and an external silicon reference: on 50 000 track projections the hit-time compression is $C = 7.0$ (X) and 11.7 (Y) with no interior term ($|b| < 0.02$), a forward model of the waveforms reproduces C to 2 %, a geometry-only model gives $C = 2.3$, and the published strip-μTPC points fall between the two. Depth-resolved stacking of the strip waveforms separates the same-time share at the mesh (equivalent to a Gaussian of 0.33–0.37 mm on the view across the resistive strips, narrower along them) from a growth with drift time that matches a transverse diffusion of $\approx 350\,\mu\mathrm{m}/\sqrt{\mathrm{cm}}$, and shows a same-time second-neighbour share of 2.4 % per side that no Gaussian footprint produces. A position-in-strip test, with its closure, bounds a pure nearest-neighbour coupling well below the published 15–30 %. The literature's remedies, scored on the same tracks against the reference, remove at most two thirds of the bias and leave the angle resolution at 1.5–2 times that of a per-event forward fit of the waveforms, which is unbiased to 1 %. We give the scaling needed to judge when a readout will see the effect and a recipe for the measurements that separate its causes.

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1 Introduction

A time-projection readout turns a drift time into a coordinate. In a large TPC the drift coordinate is read along a track many centimetres long and the local direction comes from many pad rows; in a micro-TPC (μTPC) the whole drift column is a few millimetres to a few centimetres deep, each strip crossed by the track is given one drift time, and the inclination of the segment is the slope of time against strip position [1, 2]. The second form is attractive because it recovers a good position for inclined tracks where the charge centroid degrades, and it is used in the ATLAS New Small Wheel Micromegas [3, 4, 5], in triple-GEM trackers [6, 7, 8] and in floating-strip Micromegas [9, 10].

Every one of these groups that compared the reconstructed inclination with the set angle of the chamber found the track too steep, by a few degrees at 10° shrinking to a fraction of a degree at 40° [9, 3, 8, 4]. The published explanation is capacitive coupling: a strip at the edge of the cluster receives a copy of its neighbour's signal at the neighbour's time, so the first and last strips carry the wrong depth [10, 3, 6]. The remedies are a cut on the charge ratio of the edge strips [11, 12, 13], a shift of their assigned position [7, 4], a subtraction of a fixed fraction of the neighbour pulse, sample by sample [14, 8, 15], or an angle-dependent calibration [9]. Two things are missing from this record. No study has separated the coupling from the other processes that put charge of one depth on a neighbouring strip, drift diffusion and the finite width of the induction footprint, and none has had a per-track external reference for the angle: the comparison is always with the mounting angle of the chamber, through the peak or the mean of the reconstructed distribution. Where the field's own simulations switch the coupling off they still find a bias, of 0.2° to 4° [4] or 7° [5], and leave it unexplained.

This paper argues that the question "which process couples the strips" is the wrong one for the angle. A single time per channel, whatever its definition, is a functional of all the charge that reaches the channel, and the channels at the two ends of a short column receive charge of the wrong depth from any lateral spread, by any mechanism. The bias is therefore a geometric consequence of reading a short column one time per channel, its size is set by the lateral spread in units of the pitch and by the number of channels in the column, and the mechanisms differ only in how they are to be modelled or removed. Pad TPCs and liquid-argon TPCs have the same spread in units of the pitch but columns one or two orders of magnitude longer, and read a charge-weighted time; directional and short-track detectors have the column but no reference that would reveal a few per cent of slope.

The strip μTPC is the one place where the effect is both large and measurable, and that is where it was found.

We test this account on a resistive-strip Micromegas with an unusually deep drift gap (28 mm), a fine pitch (0.78 mm), sampled waveforms (60 ns) and a silicon telescope that gives the slope of every track independently of the chamber. The detector, its calibration and its waveform reconstruction are described in two companion papers [16, 17]; here it serves as a test bed, with a forward model whose ingredients (footprint, diffusion, resistive spreading, electronics response) can be switched on and off, and with 50 000 track projections in two runs. Section 2 sets out the formulation and the scaling, section 3 what the lateral spreads are and which observables separate them, section 4 what each detector family has reported, section 5 the measurements, section 6 the remedies scored on the same tracks, and section 7 the consequences for the design and analysis of such readouts.

Terminology. We write L for the drift length of the column, p for the readout pitch, v for the drift velocity, θ for the angle between the track and the drift direction, so that a track crosses $N = L\tan\theta/p$ strips, and u for the drift time of a charge from depth $z = vu$. "Ratio" means $\tan\theta_{\rm rec}/\tan\theta$ at fixed drift velocity, which is also the factor by which the drift-time ladder is compressed; it is 1 for an unbiased reconstruction and larger than 1 when the track reads too steep. "Same-time" sharing means charge appearing on a neighbour with the time structure of the strip's own charge; "delayed" sharing means a later and longer signal, the signature of resistive spreading.

2 The single-time estimator as a functional of the track segment

2.1 What a strip sees

Consider a straight track through a drift gap of length L, inclined by θ to the drift direction, read by strips of pitch p (fig. 1a). Charge released at depth z drifts for $u = z/v$ and arrives at the readout plane at transverse position $x(z) = x_0 + z\tan\theta$, spread laterally by a distribution $f(x - x(z); z)$ of width $\sigma(z)$. The current on strip i is the sum over depths of the fraction of each depth's charge that lands within the strip, each folded with the electronics response T:

$$W_i(t) = \int_0^{L/v} du\; q(u)\, F_i(u)\, T(t - t_0 - u), \qquad F_i(u) = \int_{x_i - p/2}^{x_i + p/2} f(x - x(vu); vu)\, dx. \tag{1}$$

With no lateral spread, F_i is a box: strip i collects exactly the depths $z_i \le z < z_i + \Delta z$ with $\Delta z = p/\tan\theta$, over a time $\Delta t = p/(v\tan\theta)$. With a spread, strip i also receives charge from depths outside its box, from a range of order $\sigma(z)/\tan\theta$ on either side.

A hit-based readout keeps one time per strip, $t_i = \mathcal{E}[W_i]$, where the estimator $\mathcal{E}$ is a threshold crossing, a leading-edge fraction of the peak, the time of the peak, a fitted rising edge, a time over threshold, or a charge-weighted mean. The μTPC then fits a line $t_i = t_0' + x_i/w$ through the hits and reads the slope $w = dx/dt$ as $v\tan\theta_{\rm rec}$. For an interior strip, with charge from both neighbouring depths on both sides, every estimator returns the time of some representative depth within the strip's box, shifted by an amount that is the same for every interior strip when the spread is the same at every depth: the interior ladder keeps its slope and only its offset moves. This is the first result of the formulation and the reason a charge-weighted time, a peak time and a leading edge all give nearly the same angle: the estimator cannot change the slope of a translation-invariant ladder.

Three things break the invariance.

The partial end strips. The strip through which the track enters the gap collects charge only from $z = 0$ to the strip's far edge, a fraction ϕ of a full box; the strip through which it leaves collects the complementary fraction. Their charge is less, and their time is that of a shorter and one-sided column, while the fit assigns them the strip centre. For the entry strip the estimated time is later than the interior line extrapolated to the strip centre by about $(1 - c)(1 - \phi)\Delta t$, for the exit strip earlier by about $c(1 - \phi')\Delta t$, where c is the fraction of the column at which the estimator reads (0.3 for a 30 % leading edge, 0.5 for a centroid). Averaged over entry phases the two ends are displaced

in opposite directions by a fixed fraction of Δt. This is the "edge-strip position" effect named by Bortfeldt [10] and the "purely geometrical" bias of Ntekas's gas-only simulation [4]; it exists with no lateral spread at all.

The ghost strips. A strip whose centre lies outside the track's projection receives charge only through the lateral spread. At the mesh end its charge comes from the shallowest depths and its time is close to t_0, later than the line extrapolated to its position; at the cathode end its charge comes from the deepest depths and its time is earlier than the line. Both displacements rotate the fitted line toward smaller dt/dx, i.e. toward a steeper track. The published "ghost hit" of the capacitive-coupling literature [7, 4] is this strip; a Gaussian footprint, drift diffusion and resistive spreading produce it equally.

A spread that grows with depth. If σ grows with z the shift of an interior strip's estimated time is no longer the same at every depth. For a leading-edge estimator the earliest substantial charge on a strip comes from shallower depths through the spread, by about $\sigma(z)/(v\tan\theta)$ in time, so the deeper strips are pulled earlier by more than the shallower ones and the interior ladder itself is compressed. For a charge-weighted time the first-order effect cancels: the mean of a symmetric spread is the strip centre, and the growth of the width enters only through the gradient of σ^2, which gives a depth-independent offset $D^2/\tan^2\theta$ and no slope change [17]. Appendix B verifies both statements numerically.

2.2 The scaling

Displacing the two end points of a least-squares line through N equally spaced points by $\pm\delta$ in time changes its slope by $12\delta/(N(N+1)\Delta t)$ (appendix A). Since both the partial strips and the ghosts move the ends by amounts of order Δt (geometry) or $\sigma/(v\tan\theta)$ (spread), the ratio takes the form

$$\frac{\tan\theta_{\mathrm{rec}}}{\tan\theta} - 1 \;\simeq\; \frac{C}{N^2} + b, \qquad N = \frac{L\tan\theta}{p}, \tag{2}$$

where C collects the end displacements in units of Δt (geometry gives C of order 1 to 2; each lateral spread adds a term that grows with σ/p) and b is the interior compression from the depth-growing spread, of order $c\,D_T/(2\sqrt{z}\tan\theta)$ for a leading edge and zero for a centroid. In terms of the slope, $\Delta\tan\theta = a/\tan\theta + b\tan\theta$ with $a = C(p/L)^2$. Three consequences follow and are tested below:

1. The bias is large at small angles and falls as $1/\tan^2\theta$: the shape every strip-μTPC study reports, and the reason it was treated as a calibration. It is a property of the number of strips in the column, not of the angle as such.

2. At fixed angle it grows with the pitch, through N and through C, so coarser readout makes it worse, not better.

3. It depends on the estimator only through C and b; a change of estimator or threshold cannot remove it, and a charge-weighted time removes only the interior part b, which is small whenever the column is a few strips long.

The transparent toy of appendix B gives C a closed form for an amplitude-weighted leading-edge ladder: over thirteen configurations spanning footprints of 0.07 to 0.45 mm, diffusion on or off and pitches of 0.39 to 1.56 mm,

$$C \;\simeq\; 1.1 + 11\left(\frac{\sigma_{\mathrm{rms}}}{p}\right)^2, \qquad \sigma_{\mathrm{rms}}^2 = \sigma_0^2 + \tfrac{1}{2}D_T^2 L, \tag{3}$$

with a residual of 0.06 in C (σ_{rms} is the lateral spread averaged over the column; the spread at full drift fits six times worse). The constant is the strip geometry; the quadratic term is the ghost strips,

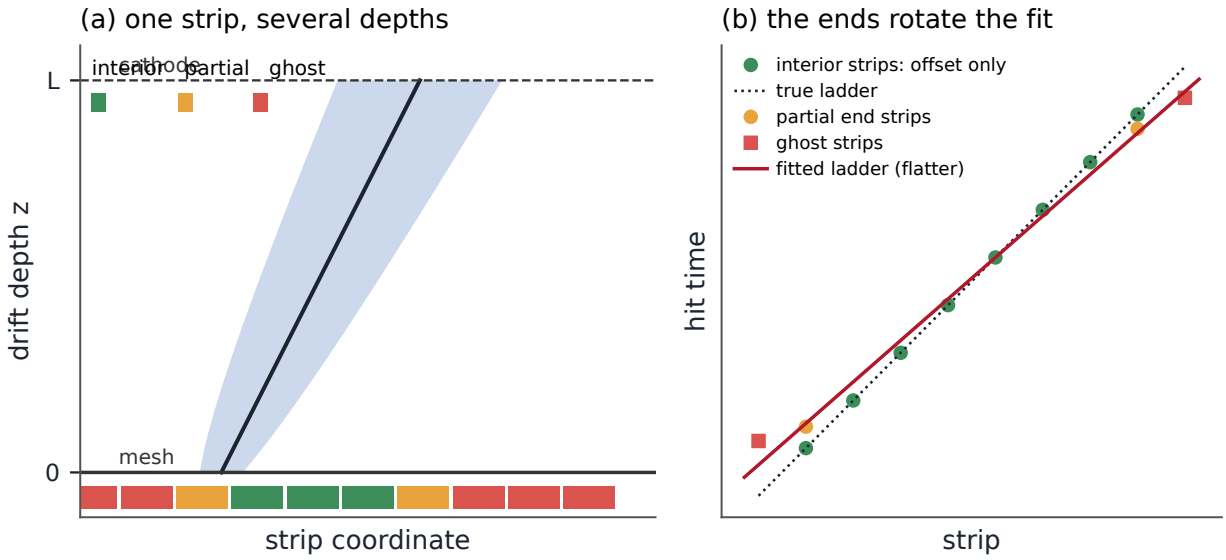


Figure 1: (a) A strip reads the charge of several drift depths: the interior strips (green) collect a full box of depths, the two partial strips at the ends of the column (orange) a fraction of one, and the ghost strips (red) only what the lateral spread (shaded) brings them. (b) In the ladder of one time per strip, the interior strips keep the true slope and move together; the partial strips and the ghosts are displaced in opposite directions at the two ends and rotate the fitted line toward a steeper track.

whose charge and whose lever arm both grow with the spread. Other estimators move C by $\pm 20\,\%$ at fixed spread (table 6).

Two dimensionless numbers therefore decide whether a readout will see the effect: N, the strips in the column, and $\sigma_\perp/p$, the total lateral spread in units of the pitch, with $\sigma_\perp^2 = \sigma_0^2 + D_T^2 L$ at full drift. The equivalent statement used in the pad-TPC literature is the lever arm $\Lambda = \sigma_T(L)/(L\tan\theta)$, the mixing width over the projected column; $\Lambda = (\sigma_T/p)/N$, so the two formulations carry the same information and eq. (2) is the more useful one because it separates the two factors.

3 What spreads the charge, and what separates the mechanisms

The literature uses "capacitive coupling" for several different processes, and a citation for one does not support another. Table 1 lists the processes that put charge of one depth on a neighbouring strip, with the time structure and the dependences that distinguish them. The last two columns are the ones that matter for the μTPC: whether the share depends on the drift depth, and whether it depends on where within the strip the charge lands.

For the angle bias the table has one message: the first five rows all feed the end strips with same-time charge, and the ladder cannot tell them apart. Within the forward model of section 5 the total compression is invariant as the same-time share is moved between a Gaussian footprint and a nearest-neighbour coupling kernel along the valley of equal waveform χ^2 [17]: a fixed fraction κ on both neighbours and a Gaussian of width σ_0 add to the cluster variance alike ($\sigma^2 \to \sigma^2 + 2\kappa p^2$), and it is the variance that sets the ghost strips. What distinguishes the processes is therefore not the bias but three other observables:

Depth. Diffusion grows as $\sqrt{z}$; resistive spreading grows with the time since landing; every same-time coupling is the same at every depth. On near-normal tracks, where the column stays within a fraction of a pitch, the neighbour share read sample by sample against the drift time separates the constant part (the same-time share at the mesh) from the growing part, and the growth measures D_T (section 5.3).

Table 1: Processes that put charge of one depth on a neighbouring strip. "Same time" means the neighbour's signal has the time structure of the strip's own charge. The MX17 column gives the values measured or bounded in section 5.

Process	Physics	Time on the neighbour	Depth dependence	Position-in-strip dependence	MX17 (chamber A)
Drift diffusion	Gaussian, $\sigma^2 = D_T^2 z$	same time	grows as $\sqrt{z}$	yes	$D_T \approx 350\,\mu\mathrm{m}/\sqrt{\mathrm{cm}}$, $\sigma_T(L) \approx 0.6\,\mathrm{mm}$
Prompt footprint	avalanche spread, charge equalisation across the resistive strip width, induction through the insulator	same time	none	yes, smooth	Gaussian-equivalent 0.33–0.37 mm (X), narrower (Y)
Floating-electrode imaging	charge on the resistive strip images onto several readout electrodes in the ratio of capacitances	same time, same shape	none	none (fixed ratio)	$\lesssim 7\,\%$ per side on X, bounded by the phase test
Direct inter-strip capacitance	strip-to-strip C into a charge amplifier: a transient leading the main pulse	derivative-like, leads by 40–115 ns	none	none	not required by the waveforms (derivative term $<$ a few % per sample)
Front-end or connector cross-talk	mutual capacitance in the ASIC or flex	same time or derivative	none	none	second neighbour 1.3 % within a chip, 0.45 % across a boundary (X)
Resistive spreading along the strip	1-D RC line, $\sigma^2 = 2D_{RC}s$ with s the time since landing	delayed and longer	grows with time since landing	yes	$D_{RC} = 3.8 \times 10^{-4}\,\mathrm{mm}^2/\mathrm{ns}$ on Y; none across the strips
Weighting-field induction	Shockley–Ramo current on a non-collecting strip	bipolar, zero net	none	yes	≈ 0 after shaping

Position in the strip. A Gaussian footprint puts most of its share on the neighbour nearer to the charge and little on the far one; a fixed coupling puts the same fraction on both. The far neighbour's share as a function of the charge's position within the strip therefore bounds the coupling (section 5.4). The test is only as good as the position estimate, which in a μTPC comes from the same waveforms; section 5.4 includes its closure.

Delay. Resistive spreading arrives later and lasts longer, which is how the T2K ERAM analysis separates its "charge spreading" from diffusion's "charge sharing" [18]. The delay separates the RC term from everything else, and nothing else: a same-time coupling and a same-time footprint sit on the same side of it.

A fourth observable, the channel map, separates electronics from the board: cross-talk inside a front-end chip or a connector follows channel adjacency, charge sharing on the board follows strip adjacency, and the two differ at chip and connector boundaries (section 5.4).

The sizes in the literature for "the" coupling fraction, 5–67 % [7, 4, 14, 5], are not one quantity. The 5 % is a pulse injected into a GEM strip and read on the neighbour after about 15 ns [7]; the 10 to 15 % are a circuit simulation and an edge-strip charge correlation on one ATLAS test chamber [4]; the 23 to 32 % are an LTspice model of resistive strips 10 to 150 cm long [14], afterwards adopted as a correction factor [8, 15]; the 67 % is the peak of an edge-to-inner charge ratio on the longest NSW strips read with the VMM, which assumes the edge strip is pure coupling [5]. A factor tuned to restore the angle, such as the 20 or 30 % of [8, 15], absorbs every same-time share, diffusion and footprint included, and measures no capacitance.

Table 2: Detector families against the two numbers that control the single-time bias: the strips in a full-drift column at $\tan\theta = 0.3$ and the transverse diffusion at full drift in units of the pitch. $\Lambda_{0.3} = \sigma_T(L)/(L\tan\theta)$ is the lever arm. Derived values rest on the cited geometry; the LMU diffusion coefficients are assumed equal to the NSW value.

Family (examples)	L [mm]	p [mm]	$N_{0.3}$	$\sigma_T(L)/p$	Time estimator; reported bias	Treatment
MX17 chamber A (this work)	28	0.78	10.8	0.75	leading edge, peak, template, centroid: ratio 1.19 (X), 1.37 (Y) at the sample median	forward fit of the waveforms
ATLAS NSW Micromegas [3, 4, 5]	5	0.425–0.45	3.4	0.6	APV25 rise time, VMM peak; 20°→23°; sharing-off simulation still +7°	edge filters, position polynomial, angle cuts
LMU floating strip [9, 10]	6	0.5	3.6	0.56	Fermi fit; 10°→15° measured, 19° in LTspice	angle-dependent rescale
LMU GEM [8]	8	0.4	6	0.82	Fermi fit; 15°→19°, 30°→32°	15–20 % neighbour subtraction
BESIII CGEM [6, 7]	5	0.65	2.3	0.38	APV25; "tilt towards larger angles"	edge weighting, 180 µm shift
T2K ERAM [18, 19]	970	10–11	27	0.3	time of maximum; RC delay, no angle bias reported	RC and gain maps; no time ladder
ALICE, STAR pad TPCs [20, 21]	2100–2500	5–7	110	0.6–0.9	charge-weighted time per row; resolution terms only	drift scale from lasers and silicon
LArTPCs [22, 23, 24, 25]	1500–3600	3–4.8	230	0.3–0.8	Gaussian peak after 2-D deconvolution; width and charge vs angle, no time bias found	deconvolution in wire and time
Directional and short-track TPCs [26, 27, 28]	10–250	0.05–2	few	0.5–8	threshold, TOT midpoint, slice barycentre; "canoe" shapes, slice ends deviate	none; resolution 1°–60°
GridPix [29]	20	0.055	110	8	time of each electron	nothing to aggregate

4 What the detector families have reported

Table 2 places the families on the two numbers of section 2.2. The first column group is the geometry, the last three are what was reported and how it was handled. All lever arms are derived from the published geometry and diffusion coefficients, since no source states the quantity; where a diffusion coefficient had to be assumed it is marked.

4.1 Strip micro-TPCs: the bias is a family trait

Every strip μTPC that published a comparison with a set angle reports the track too steep, with the excess shrinking as the angle grows. Bortfeldt measured +5° at 10° on a floating-strip Micromegas and predicted +9° with a parameter-free LTspice model of the strip capacitances, which also reproduced the fall with angle (4.3°, 2.0°, 0.6° at 20°, 30°, 40°) [9]; the journal version names two effects, "signals couple capacitively onto neighboring strips" so that "measured signal timing is shifted towards earlier values for the strips hit by electrons created close to the cathode", and "for strips at the edges of the strip cluster, the ionization electrons position is not calculated correctly" [10]. ATLAS test chambers at 20° peak near 23° [3], "understood to be due to the capacitive induction of the signal on neighboring strips", a formulation the BESIII GEM papers repeat nearly verbatim while adding that

"the transverse diffusion plays another important role" [6]. Flierl's GEM read 19° (most probable) at 15° and 32° (mean) at 30° [8]. In the units of eq. (2) these are tan ratios of 1.52, 1.17, 1.28 and 1.08 at N of 2.1, 4.5, 5.4 and 11.5 (fig. 2b), and $(\text{ratio} - 1)N^2$ lies between 2 and 11 for all of them.

The attribution rests on two roots, Bortfeldt's circuit model [9] and the NTUA/ATLAS Maxwell–Simplorer model with an edge-strip charge correlation [4]; later papers cite or restate them [11, 13, 12, 6, 7]. Neither root excludes the alternatives. Bortfeldt's model has coupling and the edge-position effect but no diffusion, and gives no decomposition; Ntekas's thesis states that the gas-only Monte Carlo "without the fully equivalent electrical circuit" still shows a bias, "purely geometrical coming from the incomplete overlap of the outer cluster strips with the particle's trace inside the MM gas gap combined with the diffusion of the ionisation electrons", of 0.2° at 30° rising to 4° at 10° [4]. Scholer's NSW digitisation, with diffusion on and charge sharing switched off, peaks 7° high for tracks above 25°, an offset he attributes to "the positions of the first and last ionizations above the corresponding strips" and leaves to "further investigation"; with sharing on the peak moves higher, and his own test beam at 28° shows no shift at all [5]. Herrmann, on a full-size NSW module, adopts a 30 % coupling correction from Lösel's simulation [14] and Flierl's analysis, and states the mechanism in strip-count terms: without the correction "no mean-time difference is in the range expected from the drift velocity", and "the time difference is biased for lower inclination due to the low number of affected strips" [15]. The only other mechanism statement we found is a 2024 conference poster by Fang et al., who derive a per-strip time correction from transverse and longitudinal diffusion on the argument that the first of several primary electrons sets the strip time [30]; it concerns the segment midpoint and has no journal version.

Two facts stand out. No source varied one parameter and tracked the bias: pitch, strip length, drift field and electronics are confounded in every comparison, and a drift-field scan changes the diffusion, the velocity, the time per strip and the signal to noise together. And no source had a per-track external angle; the set angle of a chamber is known to about a degree, which at 10° is 10 % in the tangent, the size of the effect under study.

4.2 Pad TPCs and liquid-argon TPCs: the same spread, a longer column, another estimator

The large pad TPCs have a transverse diffusion at full drift comparable to their pad width (ALICE 0.6–0.9, STAR 0.5–1.0 in units of the pad [20, 21]), the same as MX17's 0.75. What differs is the column: 45 to 150 rows against 3 to 13 strips, so the end strips are a fraction of a per cent of the points. They also read a charge-weighted time, STAR "weighting the average by the amount of charge collected in each bucket" [21] and ALICE a centre of gravity over time bins [20], which has no interior compression. Their angle-dependent terms (the angular pad and angular wire effects, $\tan^2\alpha$ in the resolution, threshold losses) are variances, not slope biases, and their drift scale is fixed by lasers and silicon to 0.1 %. A multiplicative compression of the drift coordinate would be absorbed into the effective drift velocity by a track-based calibration [20]; at the 0.1 % our scaling predicts for a 1.3 m column there is nothing to absorb.

The liquid-argon TPCs are the closest relatives of the forward model. MicroBooNE writes the wire signal as the charge convolved with a field and electronics response in wire and time and solves it as a two-dimensional deconvolution, with Wiener-type filters, because the long-range induction puts a signal on wires that collect nothing [22, 23]; ProtoDUNE-SP uses the same form [24]. Their kernel is the induced current, bipolar and calculated; the transverse diffusion stays in the deconvolved image. MicroBooNE states the mechanism for the pulse width, adjacent clouds that "begin to overlap as they spread in the yz-plane under the influence of D_T, causing additional σ_t smearing" that grows with the angle to the wire plane [31], and ICARUS tunes its wire response in angle bins [25]. A bias of the hit time or of the local slope has not been reported; with columns of hundreds of wires it would be below a per mille, and a 3 cm segment at full drift, where the lever arm is as short as MX17's, has not been tested against an external slope.

4.3 Directional and short-track detectors: the column but no reference

The keV-to-MeV recoil detectors have short tracks on fine channels and one time per channel (NEWAGE, BEAST, D3, MAIKo, MIGDAL's strip plane), or a barycentre per time slice (MIMAC, ACTAR) [26, 27, 28, 32]. Two of them describe the artefact without quantifying it: BEAST's pixel tracks take a "canoe shape" because "the pixel integration time is longer than the typical charge cloud size" [27], and MIMAC's "initial and final time slices of the track usually have a larger than average deviation from the track direction" [26]. Their angle references are kinematics at 10° to 16° resolution, source pointing with no absolute drift coordinate, or the self-consistency of the two halves of one track, which a common scale error passes. A few per cent of slope is invisible to all of them. The single-electron pixel detectors (GridPix) have nothing to aggregate and report only a time walk [29]; their drift scale is calibrated with a telescope, which would absorb a compression [33].

4.4 Drift tubes: the one time per wire is already a functional of the inclination

The oldest time-projection readout shows the general statement in another guise. The drift time of a wire is the arrival of the first electrons from the track segment seen by that cell, and its relation to the distance of closest approach depends on the track's inclination to the cell: the CMS drift tubes calibrate a maximum drift time that "depends on the track inclination and on the pattern of cells crossed" and an effective drift velocity that depends on "the inclination of the track" [34]; the BESIII drift chamber determines separate space-time relations "for 10 degree steps" of the entrance angle because the differences "grow with the increasing drift times up to as much as several hundred microns" [35]; Belle II parametrises its relations per layer, side and angle category [36]. These detectors never see the problem as a bias because the angle-dependent calibration is part of the method. The μTPC has no such calibration, because its purpose is to measure the inclination that the calibration would have to know.

4.5 What survives the survey

A transparent toy with drift diffusion, the strip geometry and a small induction footprint, and no coupling of any kind, reproduces the published biases of the NSW-type, floating-strip and GEM chambers within the uncertainty of their assumed footprints (table 7); the circuit model's predictions at 20° to 40° and the diffusion-only toy differ by less than 0.05 in the tangent ratio. The angle, in other words, never contained the information that would have identified the coupling.

The bias is not new, the model equation is not new, and the per-channel waveform fit has relatives in T2K's RC maps and in the LArTPC deconvolution. What had not been done is to measure the compression per track against an independent reference, to separate the geometric end term from the lateral spreads and the spreads from each other, and to fit the drift-depth charge profile of each track inside the convolution model. Those are the contributions of sections 5 and 6, and the companion paper [17].

5 Measurements on a deep resistive-strip micro-TPC

5.1 The test bed

The chamber is a bulk Micromegas with a 28.08 mm drift gap, 400 mm square, read by 512+512 strips of 0.78 mm pitch formed as combs of pads under a layer of resistive strips that run along the Y strips [16]. Both views are sampled every 60 ns by DREAM front-end chips, 32 samples per channel, so every strip's waveform is available offline. The gas is Ar/iC$_4$H$_{10}$ 95/5 with about 1 % water at a drift field of 350 V/cm, giving a drift velocity of 35.8 µm/ns from the column duration and 37.1 µm/ns (X) and 36.3 µm/ns (Y) from the angle scale against the reference [16]; the full column lasts 784 ns and spans $N = 36 \tan\theta$ strips. Cosmic muons are triggered by scintillators and tracked by a four-plane silicon telescope (M3) that gives each track's slope to 0.3 mm over 1.3 m, so the reference angle is independent of the chamber and known per track. Two runs on the same chamber

on consecutive nights give 12 366 and 38 364 single-track projections (an X and a Y projection per track) with a reference angle and a successful waveform fit [17].

Two readings of every projection are compared with the reference. The hit reading emulates a hit-based front end on the stored waveforms: a strip fires when a five-sample boxcar exceeds five times its noise, its time is the 30 % leading edge of the pulse (the production definition), and the angle is the amplitude-weighted straight line through the hits. Alternative estimators (time of the peak, template fit, charge-weighted mean, time over threshold, Fermi-Dirac rise) are emulated on the same samples; their logic is that of the VMM, TIGER, APV25 or SAMPA chips, at the DREAM shaping. The waveform reading is a per-event forward fit of every sample of every strip to eq. (1) with a measured impulse response, a Gaussian footprint $\sigma_0 = 0.30\,\mathrm{mm}$, drift diffusion $D_p = 0.023\,\mathrm{mm}/\sqrt{\mathrm{ns}}$ (equivalent to $D_T \approx 385\,\mu\mathrm{m}/\sqrt{\mathrm{cm}}$) and, on the Y view, resistive spreading with $D_{RC} = 3.8 \times 10^{-4}\,\mathrm{mm}^2/\mathrm{ns}$; the charge of each 60 ns depth slice is a free non-negative parameter and the track parameters (x_0, w, t_0) are searched [17]. Its angle scale is calibrated on the reference ($k_w = 1.037$ on X, 1.015 on Y); after that calibration it reads the reference slope to within 1 % at every angle (fig. 2a, W).

The same model regenerates each projection's waveforms with its own fitted track and charge profile and with one ingredient changed, and the real residual (data minus fit) added back, so that the hit emulators can be run on windows whose truth is known exactly [17]. We use the arms: *base* (the model as fitted), *metallic* (footprint reduced to the 0.07 mm induction floor and no resistive term, what a metallic readout in the same gas would show), *no diffusion*, *geometry only* (no diffusion, floor footprint, no resistive term: the strip width alone) and a footprint of 0.45 mm. The residual carries the real noise and whatever the model does not describe, so these arms include the data's unmodelled components; a transparent toy with white noise and the same template (appendix B) gives the clean physics.

5.2 The compression is a strip-end effect

Figure 2 shows the median ratio $\tan\theta_{\mathrm{hits}}/\tan\theta_{\mathrm{ref}}$ against the reference angle for the leading-edge hits of Run 1 and against N. The ratio is 4.2 at $|\tan\theta| = 0.04$, 1.55 at 0.10, 1.24 at 0.15, 1.115 at 0.22 and 1.02 at 0.35 on X, and 6.4, 1.89, 1.35, 1.17 and 1.06 on Y; Run 2 agrees within 0.02 at every angle. The fit $\Delta\tan\theta = a/\tan\theta + b\tan\theta$ gives

$$\begin{aligned} &\text{X:}\quad a = 0.0054\ (\text{Run 1}),\ 0.0056\ (\text{Run 2}), && b = -0.003,\ +0.017,\\ &\text{Y:}\quad a = 0.0090,\ 0.0088, && b = -0.019,\ +0.002, \end{aligned}$$

i.e. $C = a(L/p)^2 = 7.0$ to 7.3 on X and 11.4 to 11.7 on Y, and an interior term consistent with zero at the ± 0.02 level once the data are put on the calibrated angle scale (with the timing velocity instead, every ratio is inflated by $k_w - 1$ and b reads +0.03 to +0.05: a drift-velocity error is indistinguishable from an interior compression, which is why a μTPC cannot calibrate its own drift velocity from the ladder). The $1/N^2$ form is the one of eq. (2): on the log–log plot of fig. 2b the data lie on a line of slope -2 from $N = 1.4$ to $N = 13$. At the sample median, $|\tan\theta| \approx 0.11$ and $N \approx 4$, the ratio is 1.5 and the angle reads 2.5° to 3° too steep.

The arms decompose C (table 3). Geometry alone, the partial strips of the strip width with a point-like footprint and no diffusion, gives $C = 2.3$; the 0.30 mm footprint raises it to 3.8; diffusion with the floor footprint gives 5.4; both together give 6.9, the data's 7.0 on X. On Y the resistive spreading adds 4.6 more ($C = 11.5$ against the data's 11.7); with twice the spreading constant C would be 17. The same ingredients in the white-noise toy give 1.3, 2.8, 4.3 and 6.0 (appendix B): each ingredient contributes the same increment, and the arms sit about one unit above the toy because they carry the real residual (section 5.5).

The estimator does not change the picture. On the same planes the template-fit time (a matched filter, the best single time a strip can give) reads $C = 6.7$, the Fermi-Dirac rise 6.1, the peak time 7.2 and the threshold crossing 9.5 on X (table 4); the charge-weighted time, which the toy predicts to remove the interior term, reads 7.8 and is no better than the leading edge, because there is no interior term to remove. Nor does the pitch: summing adjacent strips into readout elements of k

Table 3: The strip-end coefficient C (ratio $-1 = C/N^2$) for the leading-edge hits: data against the reference, model arms with exact truth and the real residual, and the white-noise toy. The interior term b is given for the data; it is -0.01 to -0.02 for every arm (a 60 ns depth-quantisation artefact of the arms at steep angles) and $+0.005$ to $+0.008$ in the toy.

		C (arms)	C (toy)	data C, b
X	geometry only (strip width)	2.3	1.3	
	+ footprint 0.30 mm, no diffusion	3.8	2.8	
	+ diffusion, floor footprint ("metallic")	5.4	4.3	
	footprint + diffusion (model)	6.9	6.0 (6.5 at $D_p = 0.023$)	7.0–7.3, $b = -0.003/+0.017$
	footprint 0.45 mm + diffusion	9.1	8.0	
Y	model without resistive spreading	7.3		
	model (with D_{RC})	11.5		11.4–11.7, $b = -0.019/+0.002$
	$2 \times D_{RC}$	16.8		

pitches (fig. 3) raises a from 0.0054 to 0.0075, 0.0108, 0.0161 and 0.030 for $k = 2, 3, 4, 6$ in the data, and the geometry-only arm follows ($0.0019 \to 0.0041, 0.0075, 0.0111$), so most of the growth is the strip-end geometry itself. Fine strips are not the cause; coarser ones make it worse.

The published points of fig. 2b fall in the band spanned by the arms. Bortfeldt's measured 10° point ($(\text{ratio} - 1)N^2 = 2.3$) and Ntekas's gas-only simulation at 10° (2.0) sit on the geometry line, Iodice's NSW chamber at 20° (3.4) and Bortfeldt's circuit model (3.9 to 4.7) between the geometry and the no-diffusion arms, and Flierl's GEM (8.3 at 15°, 10.8 at 30°), whose diffusion at full drift is 0.8 of its pitch like ours, on the data line. The ordering follows the lateral spread in units of the pitch, as eq. (2) says it should; with diffusion coefficients assumed for the LMU chambers and most-probable or mean angles quoted, this is a consistency check and not a measurement.

5.3 Lateral sharing against drift depth

On tracks within 0.5° of the drift direction ($|w| < 3 \times 10^{-4}$ mm/ns, so that the track moves less than a third of a pitch over the gap) every sample of a strip's waveform belongs to a known drift time, smeared by the 200 ns impulse response. Stacking the waveforms of the five strips around the track at $t - t_0$ and dividing gives the neighbour shares against drift depth (fig. 4); the arms, regenerated on the same tracks and stacked the same way, carry the same smearing.

On the X view (569 planes) the first-neighbour share $(W_{-1} + W_{+1})/W_0$ is 0.50 ± 0.02 at t_0 and grows to 0.80 at 400 ns and 1.18 ± 0.03 at 1000 ns. With the diffusion switched off the model's share is flat at 0.46 to 0.48: the growth is drift diffusion, and its rate, read from the erf-fitted width of the five-strip pattern, is $d\sigma^2_{\text{eff}}/dt = 3.5 \times 10^{-4}\,\text{mm}^2/\text{ns}$ against 4.2 for the model's $D_p = 0.023$ and 3.2 for $D_p = 0.020$: the data prefer $D_p \approx 0.021\,\text{mm}/\sqrt{\text{ns}}$, $D_T \approx 350\,\mu\text{m}/\sqrt{\text{cm}}$, 8 % below the fitted value and inside the Magboltz range for this gas. The metallic arm starts at 0.27: without a same-time footprint of the model's size the mesh-end share is half the data's, so a same-time share is required at the mesh, equivalent in the width metric to a Gaussian of 0.33–0.37 mm on X. On Y (744 planes) the first-neighbour share grows to 2.0 by 800 ns and the resistive spreading supplies most of the growth (the arm without it reaches only 1.1); the mesh-end share is lower than on X (0.49 against 0.50 with a model that puts 0.60 there), i.e. the same-time footprint along the resistive strips is narrower than across them, as it should be if the footprint across the strips includes the equalisation of the avalanche charge over the 550 µm strip width [17].

The second neighbours tell a different story. At t_0 their share is 0.049 ± 0.004 on X and 0.047 ± 0.001 on Y, about 2.4 % per side, where any Gaussian footprint of the fitted width gives less than 0.01 and the model 0.006; the kernels fitted with a same-time second-neighbour term ($k_2 = 0.01$ to 0.02) reproduce it (0.044 and 0.061). At depth the data's second-neighbour share also exceeds every arm (0.27 against 0.24 for the model at 1000 ns on X): the lateral distribution of the collected charge has heavier tails than a Gaussian, at the mesh and at the cathode. A channel-map test [17] assigns most of the mesh-end second-neighbour share to the electronics (1.34 % within a chip against

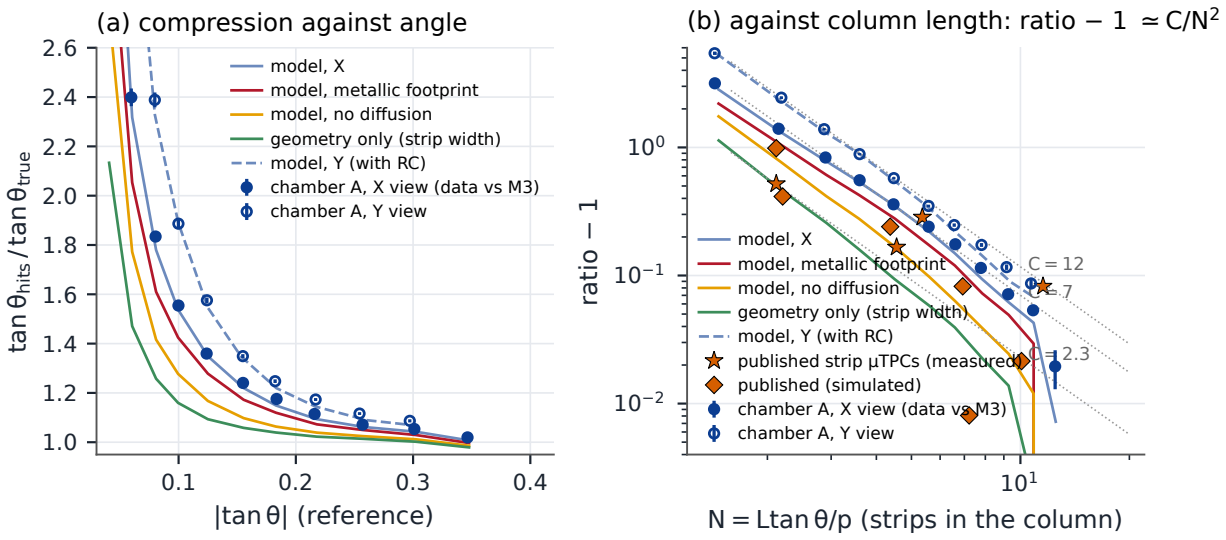


Figure 2: (a) The median ratio of the hit-time angle to the reference angle against the reference angle, for the data of Run 1 (markers; filled X, open Y) and for the model arms (lines, exact truth). (b) The same as ratio -1 against the number of strips in the column, with the published strip-μTPC points converted from the quoted set and reconstructed angles with each chamber's gap and pitch (stars: measured; diamonds: circuit or gas-only simulations). Dotted lines: C/N^2 for $C = 2.3$ (geometry), 7 (data X) and 12 (data Y).

0.45 ± 0.11 % across a chip or connector boundary on X); the first-neighbour share does not change across boundaries and is a property of the board.

5.4 Position in the strip: bounding a fixed coupling

A fixed nearest-neighbour coupling puts the same fraction on both neighbours wherever the charge lands in the strip; a Gaussian footprint puts it on the nearer one. Figure 5a shows, at t_0 on the X view, the share of the neighbour on the side the fitted track leans to ("near") and of the other ("far") in three bins of the track's position within the strip. Near the centre the two are 0.37/0.22; at the edge 0.84/0.10. The Gaussian model reproduces the near side everywhere and the far side near the centre, and under-predicts the far side at the edge (0.07 against 0.10 ± 0.01); a pure nearest-neighbour coupling of 13 % with a 33 µm footprint, the kernel that replaces σ_0 at equal waveform χ^2, is too low on the near side near the centre (0.33) and too high on the far side at the edge (0.13); a kernel with a 0.25 mm footprint and a 6 % same-time term matches all three bins on both sides. On Y the data's far share at the edge (0.12) is below every model and no same-time floor is required.

The phase is the fitted position, and a fit that leans towards the brighter neighbour could create a near/far asymmetry in symmetric data. Figure 5b closes this on 2 900 planes per view: windows regenerated with a known truth (a 0.32 mm Gaussian with no coupling, or the pure 13 % coupling with a 33 µm footprint), fresh noise at the strip's σ, and the track refitted with the production fitter exactly as for the data, are processed identically. The refitted position scatters about the truth by 0.007 mm (Gaussian truth) and 0.07 mm (coupling truth), a tenth of a pitch at most, and the near/far shares with the refitted and the true position agree within 0.01 in every bin; on the pure-coupling truth the refit turns the symmetric 0.160/0.144 of the centre bin into 0.174/0.138, a lean that falls far short of the data's 0.205/0.124. The test is therefore sound. On X the data's far share is 0.124 at the centre and then flat at 0.060 to 0.067 from a quarter pitch outward, where the Gaussian truth falls from 0.167 to 0.040 and the pure coupling stays at 0.13: the data have a floor, and it is half the published coupling. On Y the floor is 0.10 against a Gaussian's 0.07 and the coupling's 0.13, while the centre bin (0.216/0.159) lies below the Gaussian's 0.279/0.209: the same-time footprint along the resistive strips is narrower than the fitted 0.28 mm with a floor on top.

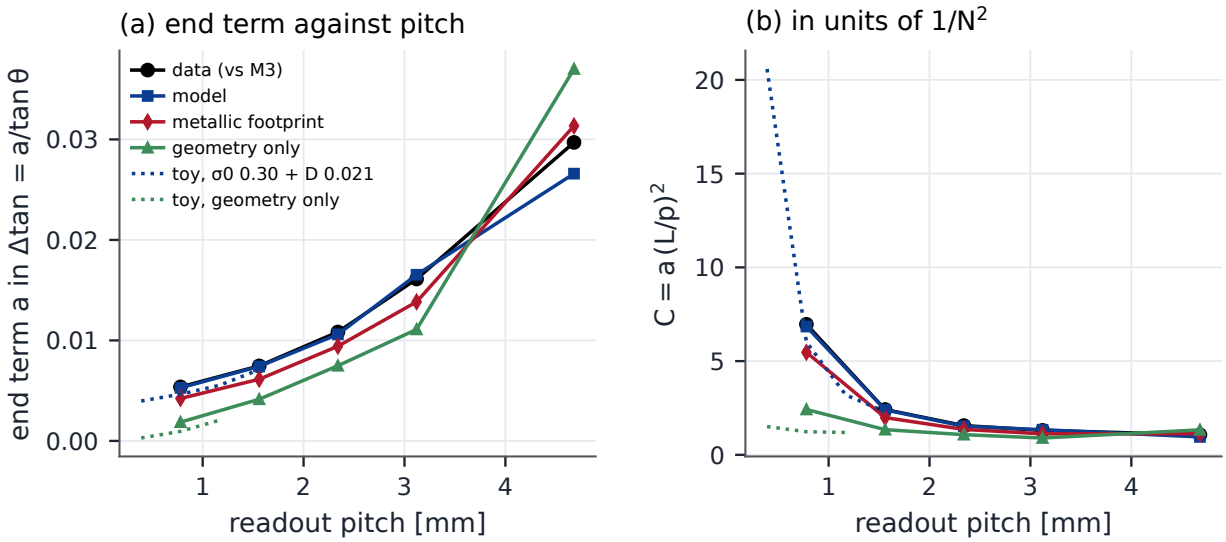


Figure 3: The end term a (left) and $C = a(L/p)^2$ (right) against the readout pitch, from summing adjacent strips of the same windows into readout elements of k pitches; data against the reference, and the model arms. The toy (dotted) is the same scan with white noise.

In both views the same-time share is a narrow core plus a flat few-per-cent floor, the shape of the kernels fitted with a footprint and a same-time term ($\sigma_0 = 0.25\,\mathrm{mm}$ with $k_1 = 0.06$ on X; 0.12 mm with 0.09 on Y [17]). A fixed nearest-neighbour coupling on chamber A is thus bounded at about 6 % (X) to 10 % (Y) per neighbour, a factor two to five below the 15 to 30 % of the strip-μTPC literature, and consistent with the published fraction that a floating resistive strip images onto the adjacent readout column for this stack-up ($\sim 1\,\%$ per side [17]) only if a further same-time process is present. Its physical origin is not identified. For the angle it does not matter: the floor's variance is what the ghost strips see, and the forward model carries it as part of σ_0.

5.5 Ghost strips

The end effect lives in the strips beyond the column. Counting the strips with a hit whose centre lies more than half a pitch outside the track's projection (fig. 6), the data have 3.4 ghosts per plane on X at $|\tan\theta| = 0.08$–0.15 against 2.6 in the model with the same noise, and the excess is at the mesh end (1.57 against 0.82 per plane; the cathode end agrees, 1.82 against 1.75). On Y the data have 5.5 ghosts per plane against 5.4 and the two ends agree within 0.3. The mesh-end excess on X is the same-time floor of section 5.3 seen from the other side: charge that no Gaussian of the fitted width brings to the second strip, at the earliest times, where it rotates the ladder most. It is the part of chamber A's compression that the model does not describe, and the reason the arms with the real residual sit one unit of C above the toy.

6 Remedies, scored on the same tracks

The literature's corrections were built for the end strips without saying so. We score each of them, and the charge-weighted time, on the same 50 000 projections against the reference, with two figures of merit: the median steepness bias in bins of the reference angle (the median of the signed angle error times the sign of the reference angle, so that the compression does not cancel between the two track directions) and the angle resolution s_{68} (the 68th percentile of the absolute error about the median within 5°) after a global linear calibration of the slope on the other half of the events. The second figure is what a μTPC with a calibration run achieves; the first is what it suffers without one. Table 4 gives both for Run 1; Run 2 agrees within 0.1 to 0.2 degrees throughout. The corrections are:

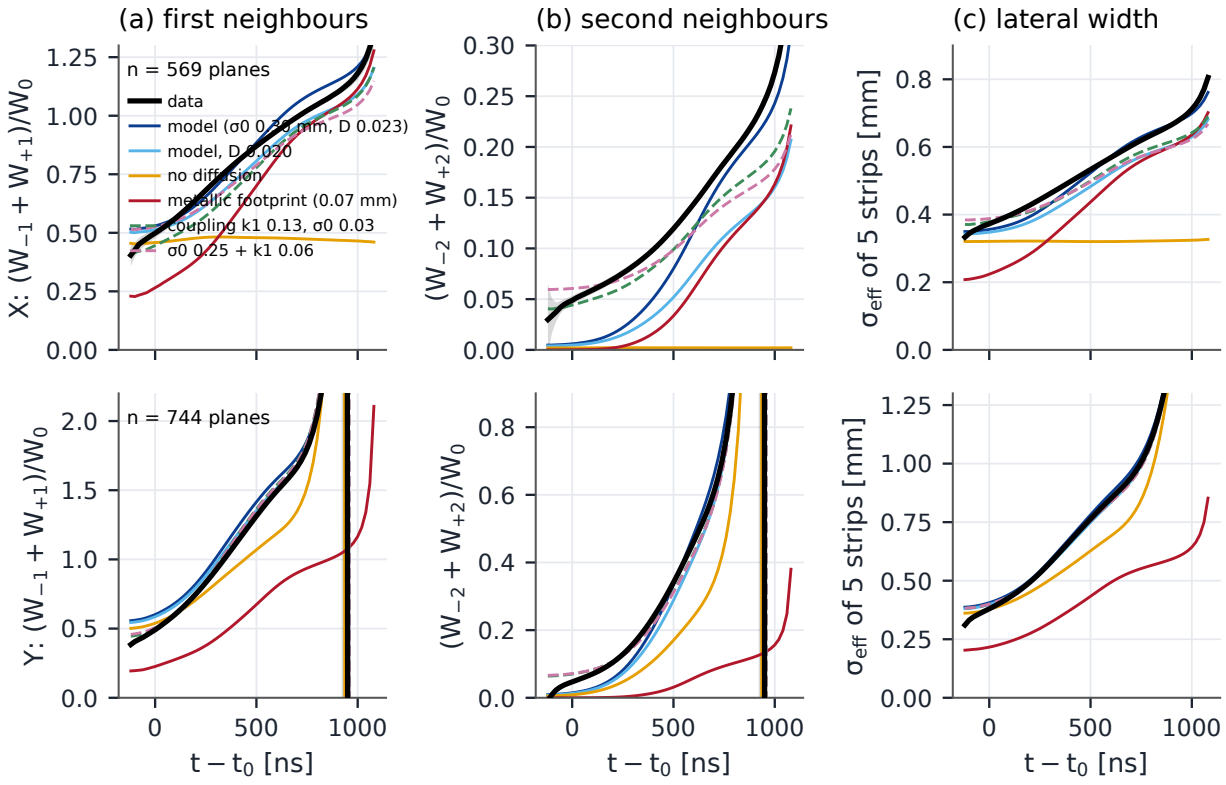


Figure 4: Lateral sharing against drift depth on near-normal tracks ($|\tan\theta| < 0.008$), X (top) and Y (bottom): (a) the first-neighbour share, (b) the second-neighbour share, (c) the erf-fitted width of the five-strip pattern, all against the time after the mesh charge, for the data (black, with bootstrap band) and the model arms regenerated on the same tracks. The growth of (a) and (c) is drift diffusion (flat without it); the mesh-end values are the same-time share; the second neighbours at t_0 are a same-time floor that no Gaussian footprint produces.

robust line iterative 3σ trimming of strips about the line, the role of the Hough step in the NSW and APV25 chains;

edge cut the first or last strip dropped when its amplitude is below 1/2 [12], 1/4 or 1/6 [11] of the inner neighbour's: the ghost-hit cut;

edge shift the kept edge strip moved inward by $(1 - a_{\mathrm{edge}}/a_{\mathrm{inner}})\,p/2$, the centroid of the partial column it collects: the geometric correction of Ntekas and Farinelli [4, 7] in a form that needs no calibration;

neighbour subtraction $W_i \to W_i - f\,(W_{i-1} + W_{i+1})$ sample by sample before the hit is formed, with $f = 0.07$ (the same-time floor of section 5.4), 0.15 [14, 8], 0.20 [8] and 0.30 [15];

charge-weighted time the centroid in time of the pulse above threshold, the estimator of the pad TPCs, which the formulation predicts to remove the interior term.

Four things follow (fig. 7).

The edge treatments act where the bias is and remove a quarter of it. Cutting or shifting the edge strips lowers the raw bias by 10 to 15 % at every angle on X and brings the calibrated resolution from 2.06° to 1.97°; the toy (appendix B) predicts a 24 % reduction of C from the edge shift alone, with the difference made up by the real chamber's ghosts, which the charge-ratio cut does not catch when they carry a few per cent of the centre strip at the same time as it.

The neighbour subtraction is the strongest single correction, and its best factor is not a capacitance. Subtracting 15 % of the neighbours' samples halves the calibrated resolution gap on X (1.69°) and Y

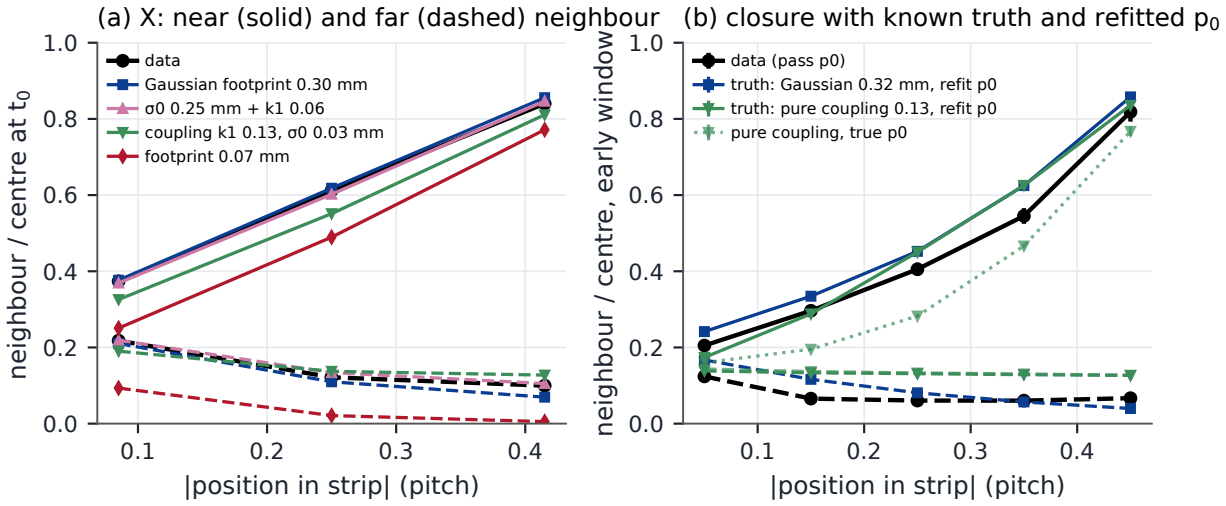


Figure 5: (a) Near (solid) and far (dashed) neighbour share at t_0 on X against the track's position within the centre strip, data and models regenerated on the same tracks with the fitted position. (b) The closure: the same quantity on windows of known truth, with the track position refitted by the production fitter.

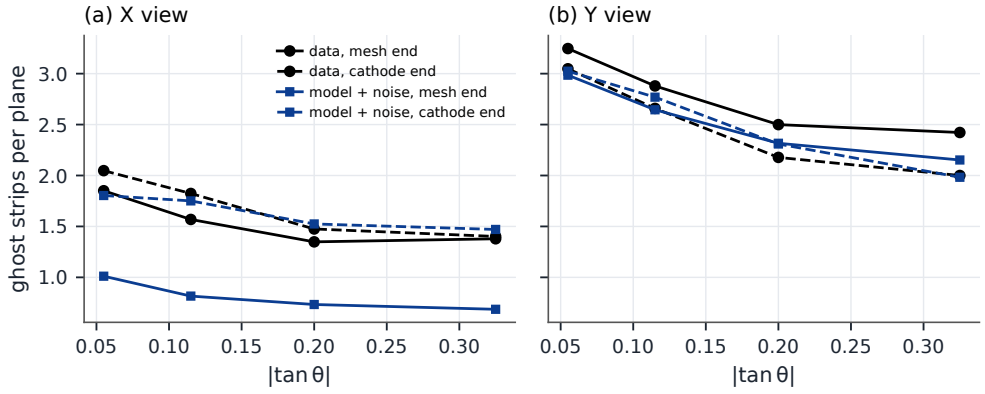


Figure 6: Ghost strips per plane (hits more than half a pitch beyond the track's projection), at the mesh end (solid) and the cathode end (dashed), data against the model with the same noise, Run 1.

(1.98°); 30 % does better on X (1.51°, bias 0.5 to 1.7°) but worse on Y (2.32°), where the neighbours' signals are the delayed resistive charge and subtracting them at the same time removes real charge. The factor that restores the angle on X, 30 %, is Flierl's and Herrmann's [8, 15], and it is four to five times the same-time coupling that the waveforms allow (section 5.4): it deconvolves every same-time share in the ladder, the Gaussian footprint and the diffusion included, and compensates the rest by over-correction. Herrmann's observation that even 40 % does not restore the expected first-to-last time difference [15] is what one expects of a nearest-neighbour subtraction applied to a spread that reaches the second neighbour and grows with depth.

The charge-weighted time does nothing by itself. It reads the same compression as the leading edge on X (2.52° against 2.48° raw) and a worse one on Y (3.05°), where the delayed resistive charge pulls a centroid later on the strips it reaches. It becomes the best estimator only after the subtraction (1.78° raw, 1.28° on Y with 15 %), because the subtraction removes the end charge and leaves an interior that the centroid reads without compression. The prediction of section 2 holds, and it is a statement about the interior term, which here is small.

No hit-level recipe closes the gap to the waveform fit. The best combination on each view, 30 % subtraction on X and charge-weighted time with 15 % subtraction on Y, leaves the calibrated

Table 4: Hit-level remedies on chamber A, Run 1, against the reference: implied drift velocity of the raw ladder (true 37.1 X / 36.3 µm/ns Y), median steepness bias per angle bin without calibration, and the angle resolution after a global scale calibration. V0 is the leading-edge hit; "template" the per-strip matched-filter time; "cw" the charge-weighted time. H2c is the forward model of section 6.1 fitted to the hit lists; W the forward fit of the waveforms [17].

view	reading	implied v [µm/ns]	steepness bias [°] at $\lvert\tan\theta\rvert$ = 0.05–0.1	0.1–0.2	0.2–0.3	0.3–0.5	s_{68} [°] calibrated
X	V0 leading edge	44.8	3.46	2.47	1.64	1.43	2.06
	V0 + robust line	44.1	3.38	2.32	1.51	1.21	2.05
	V0 + edge cut 1/2	43.4	3.09	2.12	1.36	1.08	1.97
	V0 + edge cut 1/6 + edge shift	43.6	3.16	2.16	1.42	1.15	1.97
	V0 + 15 % subtraction + robust	41.1	2.40	1.46	0.96	0.76	1.69
	V0 + 15 % subtraction + edge shift	40.3	2.02	1.25	0.82	0.74	1.54
	V0 + 30 % subtraction	39.2	1.72	0.92	0.62	0.53	1.51
	charge-weighted time + robust	44.9	3.61	2.48	1.59	1.23	2.26
	charge-weighted + 30 % subtraction	38.6	1.69	0.69	0.39	0.48	1.60
	template time + 15 % + edge shift	39.9	1.87	1.09	0.69	0.57	1.55
	H2c: forward model on hit lists			−0.05 overall			1.20
	W: forward fit of the waveforms		−0.02	0.00	−0.06	+0.02	1.04
Y	V0 leading edge	51.5	4.18	3.21	1.98	1.40	2.68
	V0 + robust line	47.2	3.85	2.50	1.43	0.94	2.42
	V0 + edge cut 1/2	46.8	3.71	2.40	1.36	0.87	2.40
	V0 + 15 % subtraction + robust	42.8	2.69	1.48	0.78	0.38	1.98
	V0 + 30 % subtraction	45.6	3.39	2.07	1.10	0.90	2.32
	charge-weighted time + robust	47.6	3.09	1.96	1.09	0.80	2.66
	charge-weighted + 15 % + robust	42.9	2.34	1.13	0.43	0.32	2.30
	template time + 15 % + edge shift	41.9	2.28	1.16	0.47	0.21	2.02
	H2c: forward model on hit lists			+0.19 overall			1.28
	W: forward fit of the waveforms		+0.03	0.00	+0.05	+0.04	1.01

resolution at 1.5° to 2.0° against 1.0° for the forward fit of the waveforms, and a residual bias of 0.5° to 2.3° without a calibration run. The forward model fitted to the hit lists alone (H2c, one time and one amplitude per strip, the physics of the waveform fit and nothing of its samples [17]) reaches 1.20°/1.28°: most of the remaining gap is information the hits do not keep, not knowledge the corrections lack.

6.1 The forward model

The alternative to correcting the hit times is not to form them. Equation (1) is linear in the charge profile $q(u)$ for a given track, so with the lateral spreads and the impulse response calibrated once per detector and run condition, the whole strip-by-sample picture of an event can be predicted for a trial (x_0, w, t_0) and the profile solved by non-negative least squares; the track parameters are then the minimum of the χ^2 over all samples of all strips [17]. The neighbours' shares, same-time or delayed, are part of the prediction and stop being contamination; the partial strips and the ghosts are predicted with the charge the model puts there and weigh in the fit by their information, not by a cut. The method is the μTPC analogue of the simultaneous multi-pad waveform fit with which T2K maps its resistive spreading [19] and of the channel-and-time convolution model of the liquid-argon deconvolution [22]; what is new is that the drift-depth profile is solved inside the fit, which is what turns a sharpened centroid into a track segment. On chamber A it reads the reference slope to 1 % at every angle (fig. 2a) with a resolution of 1.0°, against a diffusion-limited floor of 0.3° [17]; on the re-simulated arms it stays within 1 % of the truth up to ten times the resistive spreading and for every footprint and diffusion tried, because the mixing is in the model.

Its cost is the waveform: a sampled front end with enough samples to cover the column and the impulse response (here 32 at 60 ns), the calibration of the spreads per detector and run condition, and about a second of CPU per projection. A hit-only front end that keeps one time and one

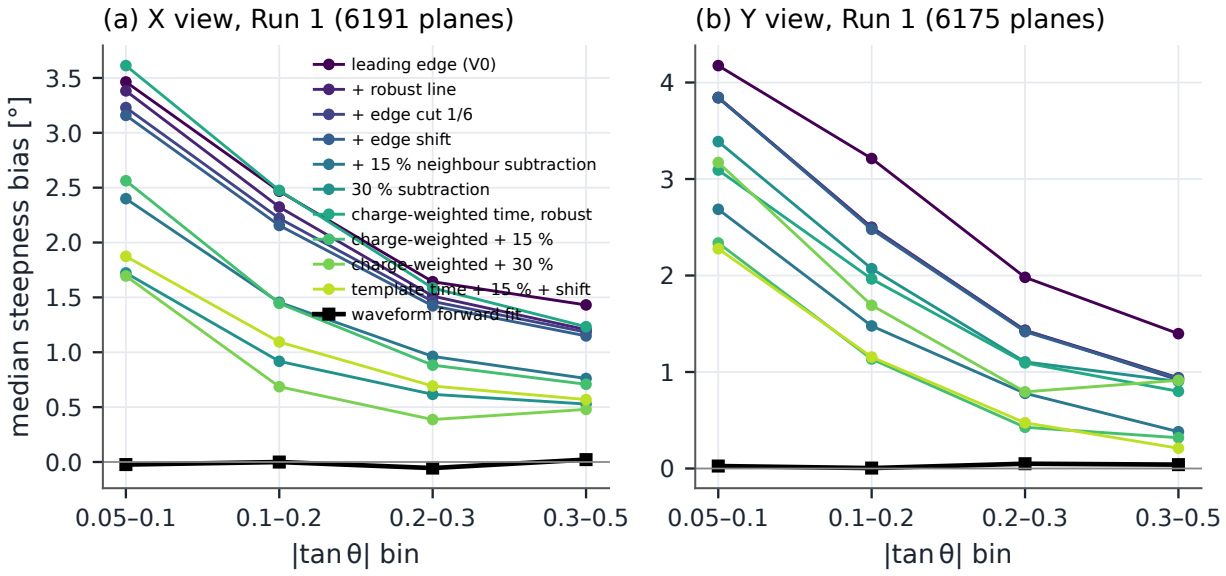


Figure 7: Median steepness bias against the reference angle for the leading-edge hits with each correction, the charge-weighted time, and the forward fit of the waveforms (black), Run 1, X (a) and Y (b).

amplitude per strip recovers most of the resolution gap only through the same model (H2c), and none of it through corrections of the hit times.

7 Consequences for design and analysis

When will a readout see the bias? Equation (2) with C from table 3 answers it. For a leading-edge or peak time and a lateral spread of 0.5 to 0.8 pitches at full drift, C is 4 to 8, and the compression is 10 % at $N \approx 7$ strips and 1 % at $N \approx 25$. A μTPC with a 5 mm gap and 0.4 mm strips has $N = 12 \tan\theta$ and reads 1.3 at 10°, 1.08 at 20° and 1.02 at 40°: the published curve. A pad TPC with a 1 m column and 1 cm pads has $N = 100 \tan\theta$ and a bias below 10^{-3} at every angle it measures; the same TPC reading a 3 cm segment near its anode would see a few per cent. A readout that sums strips to save channels raises C and lowers N at once; in the pitch scan of fig. 3 doubling the pitch raised the end term by 40 % at fixed angle.

What the hit times cannot tell you. The implied drift velocity of the raw ladder is not a drift velocity: on chamber A it reads 45 and 52 µm/ns for 37 and 36 (table 4), and every correction moves it. A μTPC that takes v from the ladder will read the right angle at one inclination and the wrong one everywhere else; one that takes v from the gas tables will read every angle too steep. A calibration against an external angle absorbs the multiplicative part and leaves the $1/N^2$ part, which no constant can remove. The interior term and a drift-velocity error have the same signature.

What to record. The remedies of section 6 rank the information: a time per strip gives the compressed ladder; a time and an amplitude allow the edge treatments and the forward model on hit lists (1.2°); the samples of the strips around the cluster allow the neighbour subtraction and the full forward fit (1.0°). Whatever is kept, the neighbours of the cluster must be kept (the VMM's neighbour-enable logic does this), because the end strips are the measurement of the spread and the model needs them. A threshold raised to suppress the ghosts removes the information that locates the column ends.

How to separate the mechanisms. The angle cannot; the compression is invariant under the exchange of a same-time footprint for a same-time coupling of equal variance (section 3). Three measurements can, and each needs only existing data: the neighbour share against drift depth on near-normal tracks (the constant is the same-time share, the growth is diffusion, a delay is resistive spreading), the far-neighbour share against the position in the strip (a floor is a fixed coupling, with its closure against the fitted position), and the channel map (electronics against board). The published coupling fractions were obtained from none of these; a fraction tuned to restore the angle is a measure of the total same-time spread, and should be reported as such.

What "capacitive coupling" should mean in the next paper. On chamber A a same-time share of a few per cent per neighbour is present, its second-neighbour part follows the channel map and is electronic, its first-neighbour part is on the board and of unidentified origin, and its contribution to the angle bias is one of four of equal standing: strip geometry, footprint, diffusion and (on the resistive view) RC spreading. A paper that attributes its μTPC bias to coupling should show the depth and phase dependences that distinguish it from the other three, or call the whole same-time spread by one name.

What the forward model needs. A measured impulse response per plane, the three spreads calibrated per detector and run condition (they trade against each other in the waveform χ^2, so the calibration must be anchored: the depth-resolved stack fixes the diffusion and the mesh-end share independently of the fit), the drift velocity from the column duration, and an angle scale from an external reference. With these it is unbiased at the per cent level and limited by diffusion and ionisation statistics [17]. Without the reference it is still unbiased in shape, which the hits are not.

8 Conclusions

The angle bias of single-time-per-channel track reconstruction, reported by every strip micro-TPC that compared itself with a set angle and attributed to capacitive coupling, is a geometric property of reading a short drift column one time per channel. A time per channel is a functional of the charge of every depth that reaches the channel; the channels at the two ends of the column receive charge of the wrong depth from the partial overlap of the track with the strip and from any lateral spread, and rotate the fitted line toward a steeper track. The ratio of the reconstructed to the true tangent is $1 + C/N^2 + b$, with N the strips in the column, C set by the lateral spread in units of the pitch and the estimator, and b an interior compression from the growth of the spread with depth that is small for a leading edge and absent for a charge-weighted time.

On a 28 mm resistive-strip Micromegas with a per-track silicon reference, $C = 7.0$ (X) and 11.7 (Y) with $|b| < 0.02$; a forward model of the waveforms reproduces C to 2 % and decomposes it into strip geometry (2.3), the same-time footprint (+1.5), drift diffusion (+3.1) and, across the resistive strips, RC spreading (+4.6); the published strip-μTPC points fall between the geometry-only and the full curves in the order of their lateral spread. Depth-resolved stacking of the strip waveforms separates the same-time share at the mesh from a growth that matches a transverse diffusion of $350\,\mu\mathrm{m}/\sqrt{\mathrm{cm}}$, and finds a same-time second-neighbour share of 2.4 % per side, mostly electronic, that no Gaussian footprint gives. The position-in-strip test bounds a fixed nearest-neighbour coupling to about 6 % on the view across the resistive strips, a factor three to five below the published fractions, and its closure shows the fitted position to be adequate for the test. Nothing in the data requires a coupling of the published size, and nothing in the angle could have told the difference.

The literature's remedies address the ends. Scored on the same tracks, the edge cuts and shifts remove a quarter of the bias, the neighbour subtraction at its tuned factor of 30 % removes two thirds on the metallic-like view and harms the resistive one, and the charge-weighted time helps only after a subtraction; the best hit-level recipe leaves the angle resolution at 1.5 to 2 times that of a per-event forward fit of the waveforms, which is unbiased to 1 % at every angle because the spreads

are in its model. The tuned subtraction factor is a measure of the total same-time spread, not of a capacitance.

For a detector builder the useful numbers are two: the strips in the column and the lateral spread in pitches. A column of more than about 25 strips, or a charge-weighted time on a column with no ghosts, hides the effect; a column of a few strips read by one time each will show it at the 10 to 50 % level, whatever the front end, and must either be calibrated against an external angle at every inclination or be read as waveforms and fitted. A μTPC cannot measure its own drift velocity from the ladder, and a coupling fraction tuned on the angle cannot be quoted as a property of the board.

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A The slope of a line with displaced ends

Let N points $x_i = i\,p$, $i = 1 \ldots N$, carry times $t_i = x_i/w + \delta_i$ and let a straight line be fitted by unweighted least squares. The slope estimate is

$$\hat{s} = \frac{\sum_i (x_i - \bar{x})(t_i - \bar{t})}{\sum_i (x_i - \bar{x})^2} = \frac{1}{w} + \frac{\sum_i (x_i - \bar{x})\,\delta_i}{p^2 N(N^2-1)/12}. \tag{4}$$

If only the two end points are displaced, $\delta_1 = +\delta$ (the mesh-end point later than the line) and $\delta_N = -\delta$ (the cathode-end point earlier), then $\sum_i (x_i - \bar{x})\delta_i = -(N-1)p\,\delta$ and

$$\frac{\hat{s} - 1/w}{1/w} = -\frac{12\,\delta}{N(N+1)\,\Delta t}, \qquad \Delta t = \frac{p}{w}, \tag{5}$$

so that the reconstructed tangent, $1/(\hat{s}v)$, is too large by

$$\frac{\tan\theta_{\mathrm{rec}}}{\tan\theta} - 1 \simeq \frac{12}{N(N+1)}\,\frac{\delta}{\Delta t} \approx \frac{12\,(\delta/\Delta t)}{N^2}. \tag{6}$$

Ghost points beyond the ends enter the same way with a larger lever arm and a displacement set by the lateral spread, $\delta \sim \sigma_\perp/(v\tan\theta) = (\sigma_\perp/p)\,\Delta t$, which gives the dependence of C on the spread in units of the pitch. Partial strips give δ a fixed fraction of Δt set by the estimator (section 2), hence a C of order unity that does not depend on the spread. Amplitude weighting lowers the weight of the partial strips and the ghosts, which carry less charge, and so lowers C; the robust trimming of section 6 lowers it further by removing some of them. The scaling with $1/N^2$ is the same for all of these, which is why the angle dependence alone does not identify the cause (section 3).

Bortfeldt's LTspice predictions [9], converted to $\Delta\tan\theta \cdot \tan\theta$, give 0.031, 0.032, 0.028 and 0.015 at 10°, 20°, 30° and 40°: a nearly constant product, i.e. the $1/N^2$ form, as expected of a circuit model that perturbs the end strips.

B A transparent toy of the single-time bias

The toy (`toy_mixing.py` in the analysis repository) implements eq. (1) with nothing else: a straight track through a gap L at $\tan\theta$, entering at a uniformly random position within a strip; ionisation in 5 ns steps of drift time, either uniform or clustered (Poisson clusters at 25 cm^{-1} with sizes distributed as $1/n^2$ and an exponential gain per electron); a Gaussian lateral spread $\sigma^2(u) = \sigma_0^2 + D_p^2 u$ integrated over strips of pitch p; the measured impulse response of chamber A's X plane (or an analytic CR–RC^2 of chosen peaking time); sampling at 60 ns (or 25 ns); white noise at a signal-to-noise of about 35 on the centre strip; the hit emulators and ladder fits of section 5. Each point is the median over

Table 5: Toy results for chamber A's geometry: the strip-end coefficient C and the interior term b for three estimators and for the leading edge with the edge shift. $D_p = 0.021\,\mathrm{mm}/\sqrt{\mathrm{ns}}$ unless stated.

	leading edge		charge-weighted		template time		leading edge + edge shift	
configuration	C	b	C	b	C	b	C	b
geometry only ($\sigma_0 = 0.07$, $D_p = 0$)	1.27	−0.001	1.33	−0.003	1.27	−0.001	0.90	−0.005
footprint only ($\sigma_0 = 0.30$, $D_p = 0$)	2.83	0.000	2.95	−0.004	2.73	−0.001	1.68	−0.001
diffusion only ($\sigma_0 = 0.07$)	4.33	+0.008	4.56	0.000	4.08	0.000	3.20	+0.006
footprint + diffusion	5.95	+0.006	6.27	−0.003	5.60	−0.001	4.53	+0.005
the same with $D_p = 0.023$	6.46	+0.007	6.81	−0.001	6.09	−0.001	5.08	+0.006
footprint 0.45 + diffusion	8.00	+0.004	8.44	−0.002	7.62	−0.002	6.55	+0.003

300 tracks. The toy reproduces the ordering and the increments of the model arms (table 3) with a constant offset of about one unit of C, which the arms owe to the real residual they carry.

Three results of the toy are used in the text. The interior term b is +0.5 to +0.8 % for the leading edge whenever the spread grows with depth and consistent with zero for the charge-weighted and template times, as section 2 derives; it is far below the end term at every N of interest. The edge shift (moving a kept edge strip inward by the fraction of the column it collects) lowers C by about 25 % for every configuration, and the charge-weighted time does not lower it at all. And C grows with the lateral spread in units of the pitch: from 1.3 with none to 2.8 with the footprint (0.38 pitches), 4.3 with diffusion alone (0.73 pitches at full drift), 6.0 with both (0.84) and 8.0 with a wide footprint (0.93); the pitch scan (fig. 3) and the other geometries of table 7 follow the same variable.

Table 6: Toy C and b for the estimators of section 6 on chamber A's geometry ($\sigma_0 = 0.30$ mm, $D_p = 0.021$), each plain and with the edge shift. The Fermi-Dirac rise and the peak time are the estimators of the APV25 and VMM chains.

estimator	C	b	C with edge shift	b with edge shift
leading edge (30 % of peak)	5.9	+0.004	4.6	+0.003
Fermi-Dirac rise, half height	4.9	+0.005	3.7	+0.004
time of the peak	5.3	−0.001	4.0	−0.002
template (matched-filter) time	5.6	−0.002	4.2	−0.003
charge-weighted time	6.2	−0.002	4.8	−0.004

Table 7: The toy on the published geometries, with no capacitive coupling of any kind: a 50 ns CR–RC^2 shaper sampled every 25 ns, a Fermi-Dirac rise-time estimator (LMU practice) or the time of the peak (VMM), $D_T = 365\,\mu\mathrm{m}/\sqrt{\mathrm{cm}}$, a footprint of 0.10 mm (0.15 mm for the GEM), clustered ionisation. The published values are those of fig. 2b.

geometry	σ_{rms}/p	C (Fermi / peak)	ratio at 10° / 15° / 20° / 30°	published
5 mm, 0.40 mm (NSW-like)	0.52	4.1 / 4.9	1.82 / 1.46 / 1.27 / 1.10	20°→23°, ratio 1.17 [3]
6 mm, 0.50 mm (floating strip)	0.45	3.0 / 4.0	1.68 / 1.35 / 1.19 / 1.06	1.52 measured at 10°; 1.99 / 1.24 / 1.08 at 10° / 20° / 30° in LTspice [9]
8 mm, 0.40 mm (GEM)	0.69	5.9 / 6.8	1.50 / 1.22 / 1.13 / 1.05	1.28 at 15°, 1.08 at 30° [8]
28 mm, 0.78 mm (chamber A, X)	0.66	6.4 (leading edge)	1.16 / 1.08 / 1.04 / 1.02	data 1.17 / 1.08 / 1.04 at $N = 6.3$ / 9.6 / 13 (section 5.2)

The last table is the central check of the review's argument: the biases the strip-μTPC groups measured and simulated, 1.1 to 2.0 at 10° to 30°, are reproduced within the uncertainty of the assumed footprint by drift diffusion, the strip-end geometry and a 0.1 mm footprint, with no coupling term. The circuit model's 20–40° predictions (1.24, 1.08, 1.02) and the diffusion-only toy (1.19, 1.06,

1.03) are indistinguishable at the precision of either; at 10° the measured 1.52 lies between them. Coupling may be present in those chambers, but the angle did not need it.

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